Distilling in the Dark - Effective Knowledge Distillation in LLMs Under Data Constraints

Kaan Nadir Sirin

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Lund University

Faculty of Engineering
Centre for Mathematical Sciences
Mathematics

CENTRUM SCIENTIARUM MATHEMATICARUM



LUND UNIVERSITY

LTH

FACULTY OF ENGINEERING

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Abstract

Large language models (LLMs) have achieved remarkable capabilities but remain costly to deploy at scale. Knowledge distillation offers a way to transfer knowledge from large, high-capacity models into smaller, more efficient ones. This thesis investigates two key questions: (1) how specific design choices affect distillation performance, and (2) how distillation can be effectively performed in the absence of high-quality datasets. The first part of the thesis presents a systematic evaluation of distillation parameters. We show that forward KL divergence with a temperature of $T = 1$ consistently yields stronger student models than reverse KL or higher temperatures. Scaling the number of training examples further improves performance, and distilled models can match or outperform their undistilled counterparts on out-of-domain tasks. In the second part, we introduce a novel distillation method based on generation and sparse teacher logits. This approach enables effective knowledge transfer with as few as 320 examples and outperforms supervised fine-tuning. Among sparsification strategies, importance sampling proves superior to top-K even when using sparser distributions. Together, these results outline a practical and data-efficient strategy for distilling large models, even in settings with limited and low-quality data.

Keywords: large language models, knowledge distillation, sparse logits, importance sampling, low-resource learning

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1 Introduction

Large language models (LLMs) have reshaped our understanding of what it means to extract knowledge from plain text. Yet the very scale that allows these models to learn rich representations often makes them impractical for deployment, requiring expensive hardware or introducing latency.

Knowledge distillation is one way to bridge this gap. It relies on the idea that a large model, once trained, may contain redundant complexity. Knowledge, once extracted, can often be represented much more compactly. Indeed, many leading research labs—such as Meta with Llama, Google with Gemma, and OpenAI with GPT-4o-mini—use distillation to transfer the essence of expensive and cumbersome models into more efficient counterparts [4, 36, 13].

Still, it's often unclear how these labs apply distillation in practice. This thesis takes an exploratory look at knowledge distillation in LLMs and introduces a novel approach that works even in the absence of high-quality datasets. The goal is to gain a deeper understanding of the process and expand its applicability to new, underexplored settings.

1.1 Research Objectives

The thesis is organized around the following two lines of investigation.

1.1.1 Design Choices Influencing Distillation Performance

The first part of the thesis explores how specific design choices impact the efficacy of knowledge distillation. Here, we systematically vary key distillation parameters and evaluate how each choice affects the performance of the resulting student model.

1.1.2 Distilling in the Dark

One major challenge in distillation is the lack of suitable datasets. In the second part, we introduce a new technique that enables distillation even in cases where good datasets are unavailable.

We first demonstrate the effectiveness of this approach with a toy example, where we inject a specific style into a student model via distillation. We then compare the results of our method to supervised fine-tuning (SFT).

After establishing its superiority over SFT, we apply the technique in a more practical setting: distilling a model on a Swedish medical multiple-choice QA dataset, a format that is generally ill-suited for distillation due to its short and often uninformative one-letter answers.

1.2 Research Questions

This thesis aims to answer the following research questions:

RQ1 — How do specific design choices (e.g., sampling strategies, loss functions, and temperature settings) influence the performance and effectiveness of knowledge distillation in large language models?

RQ2 — How can knowledge distillation be effectively performed when high-quality datasets embedding the target knowledge are unavailable or insufficient?

2 Background

This section introduces key concepts and terminology that form the foundation for the rest of this study.

2.1 Large Language Models (LLMs)

Large language models are large scale neural network models trained on massive text corpora. These models often feature several billions of parameters allowing them to capture statistical patterns in language, and thereby letting them perform a wide array of tasks across domains.

As LLMs scale up, they not only improve in accuracy but also begin to exhibit emergent abilities—capabilities not present in smaller models. A landmark demonstration of this phenomenon was GPT-3, an autoregressive language model with 175 billion parameters, over ten times larger than any previous non-sparse model at the time. In the few-shot setting, GPT-3 was able to achieve competitive performance with prior state-of-the-art fine-tuned models, without task-specific training [7].

LLMs can be grouped into three main architectural categories, encoder-only models (e.g. BERT), encoder–decoder models (e.g. T5), and decoder-only models (e.g. GPT-4, LLaMA). Decoder-only models are primarily used for generative tasks like text completion and dialogue. In this study, the focus will be exclusively on decoder-only architectures [14, 34, 1, 37].

2.1.1 Decoder-only Architecture

Decoder-only language models are based on the Transformer architecture introduced by Vaswani et al. [39], which introduced the self-attention mechanism. This

mechanism enables models to capture long-range dependencies and contextual information more effectively than previous architectures.

Decoder-only models, as the name suggests, consist only of the Transformer decoder stack. They are typically trained for autoregressive generation, that is, to predict the next token given all previous tokens and can later be fine-tuned or prompted for specific tasks [27].

This simple, unified setup was pioneered by Radford et al. [32], later scaled up and shown to yield consistent performance gains as model size increases [33, 7, 25].

2.1.2 Tokens and Vocabulary

In the context of large language models (LLMs), a token refers to the smallest unit of text the model processes. Tokenization is the preprocessing step that breaks input text into tokens, which may be entire words, subword fragments, or individual characters, depending on the tokenizer. Each token is then mapped to a unique integer ID. Modern LLMs typically use subword tokenizers [24].

For example, using the LLaMA 3 tokenizer—the model class used in the experiments throughout this thesis—the sentence *"LLMs are great"* is tokenized as `['LL', 'Ms', ' are', ' great']`[1]. These tokens correspond to the integer IDs `[4178, 22365, 527, 2294]`.

The vocabulary is the set of all tokens that the model can recognize or generate. In other words, it is the fixed collection of unique tokens defined during training. LLaMA 3 uses a vocabulary size of 128,256 tokens [3].

2.2 Knowledge Distillation

Knowledge distillation is a technique for transferring the knowledge and capabilities of a large *teacher* model to a smaller, more deployment-friendly *student* model.

The idea of training a smaller model to mimic the function learned by a more complex model was first introduced by Caruana et al. [8], who trained a compact neural network to approximate a function learned by ensemble selection. This

[1]The actual tokens are `['LL', 'Ms', 'Ġare', 'Ġgreat']`, where the symbol Ġ indicates a leading space.

was achieved by using the ensemble to label a large unlabeled dataset and then training a neural network on the resulting "soft labels" (i.e., probabilistic outputs) yielding a model that outperformed one trained on the original training set.

Their work demonstrated that training on the soft outputs of a teacher model could encode more information than hard labels; an idea that later became central to Hinton et al.'s seminal work, where the term knowledge distillation was coined. Hinton et al. further extended this concept by applying it within deep neural networks, introducing a temperature parameter to soften the teacher's softmax outputs—thereby revealing richer class relationships—and proposing a combined loss function that blends soft target imitation with supervision from hard ground truth labels [18].

Today, knowledge distillation techniques are commonly categorized into three primary types: response-based, feature-based, and relation-based distillation [40].

- **Response-based distillation**, also referred to as *logit-based disillation*, transfers the soft output probabilities (logits) from the teacher to the student, allowing the student to mimic the teacher's final predictions. This is the approach described in Hinton et al.'s original paper.
- **Feature-based distillation** aligns intermediate representations or feature maps between the teacher and student, facilitating the transfer of learned features. This approach is also known as *hidden-state-based distillation*.
- **Relation-based distillation** captures and transfers structural relationships between data samples or features, such as pairwise distances or angles.

Among these three approaches, response-based distillation is reported as the most widely adopted [21].

An empirical study by Arcee AI compared the response-based and hidden-state based approaches on standard LLM benchmarks and found that both approaches surpassed a supervised-fine-tuned (SFT-only) baseline on BBH, MUSR and MMLU-PRO. However, the logit variant was found to deliver stronger overall results compared to the hidden-state-based approach [2].

Motivated by these findings, its straightforward yet effective nature, and its widespread use, the remainder of this thesis focuses on logit-based distillation for generative LLMs.

2.2.1 Logit-based Distillation in LLMs

Logit distillation refers to the standard form of knowledge distillation where the student model is trained to match the teacher model's output logits i.e. the probability distribution over the vocabulary in the context of LLMs. In other terms the teacher's output distribution acts as soft targets. Instead of training the student solely on ground-truth labels, the student learns from the teacher's output distributions (the "soft" targets), which contain richer information than hard labels alone.

Typically the loss used in logit-based knowledge distillation is a weighted average of the Kullback–Leibler Divergence (KLD) between the student and teacher distribution and the cross entropy loss for student predictions over the hard targets, ie. the input tokens.

Kullback–Leibler divergence (KLD)

Kullback–Leibler divergence (KLD) is a measure of how one probability distribution diverges from another reference distribution [31].

$$D_{\mathrm{KL}}(P \parallel Q) \;=\; \sum_{x \in \mathcal{X}} P(x)\, \log \frac{P(x)}{Q(x)}. \tag{2.1}$$

KLD is always non-negative and is zero if and only if the distributions are identical.

Furthermore, KL divergence is asymmetric:

$$D_{KL}(P \parallel Q) \neq D_{KL}(Q \parallel P) \tag{2.2}$$

In the knowledge distillation literature, a distinction is often made between *forward* and *reverse* KLD [16]. The former refers to the left-hand side of the inequality in (2.2) and is given by Equation 2.1.

The right-hand side of the inequality in (2.2) is referred to as *reverse KLD* and is defined as:

$$D_{\mathrm{KL}}(Q \parallel P) \;=\; \sum_{x \in \mathcal{X}} Q(x)\, \log \frac{Q(x)}{P(x)}. \tag{2.3}$$

The practical difference between the two lies in what they penalize. Forward KLD encourages the student to match the teacher's distribution where it places probability mass, while not strongly penalizing the student for assigning probability to outcomes the teacher considers unlikely.

In contrast, reverse KLD penalizes the student whenever it assigns probability to outcomes that the teacher assigns low probability. This leads to mode-seeking behavior: the student focuses more narrowly on the peaks of the teacher's distribution and avoids its low-probability regions [38].

It is possible to find studies reporting better performance with reverse KLD [16], as well as others favoring forward KLD [30].

Temperature scaling

Before computing the Kullback–Leibler divergence, both teacher and student logits are typically passed through a temperature-scaled softmax. Given logits $\mathbf{z}$, applying temperature scaling with a temperature $T > 1$ yields:

$$\sigma_i(\mathbf{z}; T) = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}. \tag{2.4}$$

Temperature scaling produces a softer probability distribution over classes by reducing output confidence, thereby revealing the relative differences between the non-argmax classes, as illustrated in Fig. 2.1.

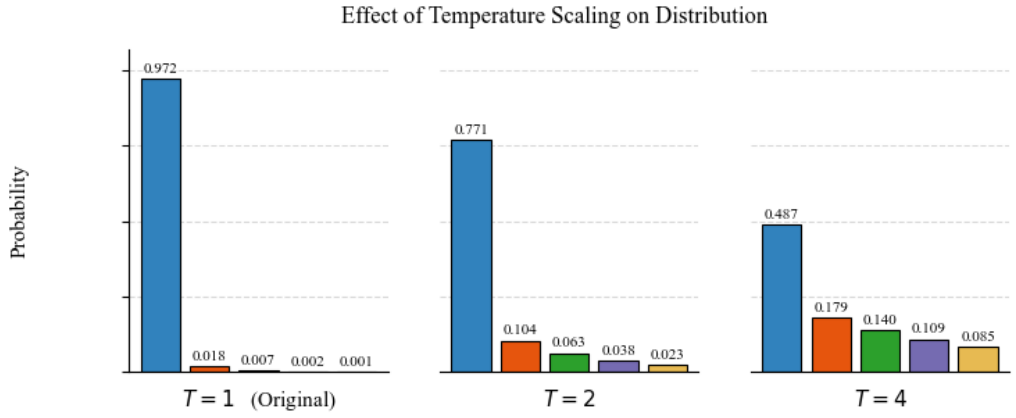


Figure 2.1: Effect of temperature scaling on the softmax distribution.

In the original distillation framework proposed by Hinton et al. [18], both the teacher and student use a high temperature during training to ensure that the

student learns from these softened outputs. After training, the student reverts to using a standard temperature of $T = 1$ for inference.

When available, ground-truth labels can also be incorporated alongside the soft targets by using a weighted sum of two loss terms: (1) the cross-entropy with the teacher's soft targets, computed at high temperature, and (2) the cross-entropy with the hard labels.

Combining the two objectives yields the final distillation loss

$$\mathcal{L}_{\mathrm{KD}} = \alpha\, T^2\, D_{\mathrm{KL}}\big(\sigma(\mathbf{z}_t; T) \,\big\|\, \sigma(\mathbf{z}_s; T)\big) \;+\; (1-\alpha)\,\mathrm{CE}\big(\mathbf{y}, \mathbf{z}_s\big) \tag{2.5}$$

where

- $\mathbf{z}_s$ and $\mathbf{z}_t$ are the student and teacher logits, respectively;
- $\sigma(\mathbf{z}; T)$ is the temperature-scaled softmax from Eq. 2.4;
- D_{KL} denotes the (in this case forward) Kullback–Leibler divergence;
- CE is the standard cross-entropy with the hard labels $\mathbf{y}$;
- $T \geq 1$ is the distillation temperature;
- $\alpha \in [0, 1]$ controls the trade-off between soft-target imitation and hard-label supervision.

The factor T^2 ensures that the gradient magnitudes from the soft-target term remain comparable to those from the hard-label term, preserving their relative influence even as T varies [18].

2.2.2 Capacity Gap

Effective distillation requires an appropriate match between teacher and student model capacities. When the teacher is too large relative to the student, the performance of the distilled model can degrade. This problem is known as the *capacity gap*. Mirzadeh et al. [28] formally characterize this issue and introduce *teacher assistants*, intermediate-sized models that bridge the capacity gap between teacher and student models.

Zhang et al. [9] further demonstrate that student loss plotted against teacher cross-entropy follows a broken power law: initially, stronger teachers reduce student loss, but beyond a certain point additional teacher improvements worsen it.

In the same work, they conduct a large-scale temperature sensitivity analysis and find that, unlike in classification tasks, the optimal distillation temperature for language modeling is consistently $T = 1$. They show that $T = 1$ consistently yields the lowest student cross-entropies across all model scales and teacher–student size ratios, from models of equal size up to over 39× difference. They attribute this to the nature of language modeling, where outputs are already soft and encode complex ambiguity. Thus, preserving the teacher's original probability ratios (i.e., using $T = 1$) best maintains this ambiguity structure [9].

2.3 Evaluation Methods

2.3.1 Perplexity

Perplexity is a long-standing metric for evaluating language models, measuring how well a model predicts a sequence of tokens [22]. Formally, for a held-out sequence of N tokens $\{w_1, \ldots, w_N\}$, the model's perplexity is defined as:

$$\mathrm{PP}(\theta) = \exp\left(-\frac{1}{N}\sum_{i=1}^{N}\log p_\theta(w_i \mid w_{<i})\right) \tag{2.6}$$

Lower perplexity implies that the model assigns higher likelihood to the observed data. However, Holtzman et al. [19] show that models optimized solely for low perplexity often produce degenerate text (e.g. repetition, low diversity), indicating that perplexity correlates poorly with generation quality in open-ended tasks and highlighting the need for complementary evaluation methods.

2.3.2 LLM-as-a-Judge

For generation tasks like summarization or dialogue, where outputs are open-ended and ground truths are ambiguous, standard metrics such as BLEU or ROUGE often fall short. A recent alternative is *LLM-as-a-Judge* [44], where a powerful language model (like GPT-4o[2]) is prompted to evaluate candidate outputs using a predefined criteria; assessing aspects such as factuality, coherence, or style.

This evaluation paradigm has been adopted in several benchmarks, including

[2] `https://openai.com/index/hello-gpt-4o/`

MT-Bench [6], and used in model comparisons by projects like Vicuna [11]. When prompted properly, LLMs have been shown to align closely with human judgment, enabling large-scale and cost-effective evaluation.

Some known limitations in the LLM-as-a-judge setup include **position bias**, where the model tends to favor the first response; **verbosity bias**, where longer outputs are rated more positively regardless of their actual content; and **self-enhancement bias**, where the judge overestimates the quality of its own responses [44].

2.4 Related Work

Recent work in knowledge distillation has expanded beyond the original formulation introduced by Hinton et al. [18], exploring new strategies to enhance distillation effectiveness, data efficiency, and model compatibility. This section reviews related work that introduce targeted distillation techniques and propose improvements in logit sampling strategies, work that has directly inspired the development of the novel approach proposed in this thesis.

2.4.1 Targeted Knowledge Distillation Techniques

Several studies deviate from the standard formulation of knowledge distillation to enable the transfer of specific types of knowledge or to improve data efficiency. Hsieh et al. [20] introduce a method called *Distilling Step-by-Step*, in which large language models (LLMs) are used to generate intermediate rationales (reasoning steps) for each task. These rationales are then used to train smaller models in a multi-task setup that combines label prediction with rationale generation. By exposing the reasoning behind predictions, rationales convey richer task-specific knowledge that is often difficult to extract from input–label pairs alone. The authors demonstrate that this method consistently outperforms standard finetuning, even with significantly fewer labeled examples, highlighting its strong data efficiency.

In a related effort, Yang et al. [41] propose a method known as *rule distillation*, which focuses on explicitly teaching structured rules to student models. Unlike the conventional learn-from-examples paradigm that often requires large volumes of annotated data, this approach aims to encode complex task-specific rules into model parameters using minimal supervision. The method is evaluated across three tasks: a synthetic arithmetic task involving a novel operator, a safety task

that trains models to resist jailbreak-style adversarial prompts, and a sentiment steering task governed by fine-grained conditional rules. Instead of relying solely on in-context learning the authors extract internal hidden-state signals from a *rule-aware* teacher LLM and use these as supervision to train the student. Experimental results show that this paradigm enables faster and more robust acquisition of task-specific rules compared to traditional example-based learning, particularly in low-data settings.

2.4.2 Sparse-logit Techniques in Knowledge Distillation

This section provides an overview of two methods for sparsifying logits in offline distillation with precomputed teacher outputs: *top-K sampling*, which retains only the k most likely tokens, and *importance sampling*, which approximates the full distribution via reweighted samples.

Top-K sampling

In top-K sampling, only the K highest logits are retained and renormalized using softmax; the rest are discarded. While this reduces memory usage, it introduces significant bias into the teacher signal by distorting the teacher's output distribution. Tokens outside the top K are assigned zero probability, even if the teacher considered them plausible, while the retained logits are rescaled to sum to one, exaggerating their relative importance.

Empirically, using fewer than 25 tokens in top-K distillation results in worse student performance than not distilling at all. Performance only approaches that of full-logit distillation when around 300 tokens are retained per step.

This issue is also evident in gradient alignment. Compared to full-logit distillation, the gradients from top-K distillation diverge markedly. For example, with $K = 12$, the angle between gradients reaches 58°, with a norm mismatch of 2.4×. Even at $K = 300$, the gradient angle remains 30° [43].

These drawbacks are so pronounced that releasing only top-K logits has been proposed as a defense against model theft: it allows the teacher model to remain functional while significantly degrading the performance of any downstream distillation attempts, sometimes leading to drops of over 20 percentage points [26].

Importance-sampling

Importance sampling offers an unbiased alternative to top-K sampling. Instead of retaining only the top logits, a small number of tokens are sampled from a softened proposal distribution defined as $q_i \propto p_i^\tau$, where p_i is the softmax probability and τ is a temperature parameter. Each sampled token is reweighted by $w_i = \frac{p_i}{q_i}$, producing an unbiased estimator of the original distribution.

This method yields gradients that closely match those from full logit distillation. Even with only 12 sampled tokens, the angular difference between gradients is as low as 4°, with norm alignment. Furthermore, Zhou et al. [43] show that using 12 tokens per step achieves comparable performance to full-logit distillation in both training loss and downstream accuracy, while being 2.6× faster and requiring 100× less storage.

3 Methodology

This chapter outlines the methods used to carry out the distillation experiments, which consist of two main parts. In the first part, we perform standard logit distillation and investigate how different distillation parameters affect the performance of the student model, as well as how well it generalizes the capabilities learned during distillation to tasks that differ from those seen in training. In the second part, we introduce a novel approach that enables distillation in settings where data is limited in quantity and suboptimal in quality or format.

3.1 Models, Datasets and Frameworks

This section describes the models, datasets and the frameworks used in the distillation experiments.

3.1.1 Models

The experiments make exclusive use of models from Meta's Llama 3 family[1], specifically the following instruction-tuned variants:

- LLaMA 3.2 1B[2]
- LLaMA 3.2 3B[3]
- LLaMA 3.1 8B[4]
- LLaMA 3.3 70B[5]

None of the models are trained from scratch; instead, student models are distilled

[1] https://ai.meta.com/blog/meta-llama-3/
[2] https://huggingface.co/meta-llama/Llama-3.2-1B-Instruct
[3] https://huggingface.co/meta-llama/Llama-3.2-3B-Instruct
[4] https://huggingface.co/meta-llama/Llama-3.1-8B-Instruct
[5] https://huggingface.co/meta-llama/Llama-3.3-70B-Instruct

from pretrained checkpoints.

Models from the same family are used because they share an identical vocabulary, both in terms of size and tokenization. This simplifies the distillation process, as it eliminates the need for additional components such as vocabulary-mapping layers or embedding projections that would otherwise be necessary when distilling between models with different token sets.

All models are used in their instruction-tuned form, meaning they have undergone supervised fine-tuning on datasets consisting of instruction-response pairs. This fine-tuning process aligns the models' outputs with human preferences and makes them better suited for following natural language instructions.

3.1.2 Datasets

Three primary datasets were used for distillation:

- **Magpie Swedish**[6]: A filtered dataset containing roughly 200,000 instruction-response pairs in Swedish. The data was cleaned by deduplication, quality filtering based on ArmoRM-LLaMA3-8B-v0.1 scores, and by limiting sequence lengths to between 10 and 2048 tokens. The dataset includes a mix of general, mathematical, and coding instructions.
- **MedQA-SWE**[7]: A multiple-choice clinical question-answering dataset in Swedish comprising 3,180 manually curated questions derived from Swedish medical exams [17]. It serves as a benchmark for evaluating factual knowledge in the medical domain.
- **Alpaca**[8]: A 52,000-sample dataset of instruction-following pairs generated using OpenAI's `text-davinci-003`. This dataset was used for style-injection experiments. Only the instruction texts were used; the corresponding responses were generated by the teacher model.

In addition to these three datasets, three publicly available benchmarks were used exclusively for evaluation.

[6] `nicher92/magpie_llama70b_200k_filtered_swedish`
[7] `nicher92/medqa-swe`
[8] `tatsu_lab/alpaca`

These benchmarks were selected to test generalization across domains and tasks not seen during training:

- **MBPP**[9]: The Mostly Basic Python Problems dataset contains Python programming problems with input/output tests and reference solutions [5]. It is used to evaluate code generation capabilities.
- **HumanEval**[10]: A dataset of 164 hand-written Python programming problems, also designed for evaluating the functional correctness of code generated by language models [10].
- **GSM8K**[11]: A dataset of 8,500 high-quality grade-school math word problems, designed to assess arithmetic and reasoning capabilities [12].

3.1.3 Frameworks

The following frameworks were used to support model distillation and evaluation throughout this work:

DistillKit[12] is an open-source toolkit developed by Arcee AI for large language model distillation. It supports both logit-based and hidden state-based distillation [2]. For this work, the toolkit was forked, restructured, and significantly extended. Enhancements include added plotting, support for reverse KL divergence, and the implementation of a novel custom distillation pipeline tailored to the experiments conducted in this thesis.

LM Evaluation Harness, developed by EleutherAI, provides a standardized evaluation framework for evaluating generative language models across a widely used academic benchmarks [15]. In this project, the harness was used to evaluate the distilled models on HumanEval, MBPP, and GSM8K.

3.2 Experimental Design

This section outlines the experimental setups used to explore different forms of knowledge distillation, including both standard logit-based methods and novel approaches involving sparse teacher distributions.

[9] `google-research-datasets/mbpp`
[10] `openai/openai_humaneval`
[11] `openai/gsm8k`
[12] `https://github.com/arcee-ai/DistillKit`

Throughout all experiments, a random 10% validation split was used, with a batch size of 2 and gradient accumulation over 8 steps. Training was run for three epochs using bf16 precision. The learning setup included a learning rate of 2×10^{-5}, weight decay of 0.05, a warmup ratio of 0.1, and a cosine learning rate scheduler. These choices were made to keep memory usage and runtime low, enabling quick experimentation. Early stopping was applied if the validation loss failed to improve for one full epoch. As the goal of this work was to explore distillation strategies rather than optimize hyperparameters, no systematic tuning was performed.

3.2.1 Standard Logit Distillation

In the first experiment, a student model (`3B`) was distilled from a larger teacher model (`8B`) using the Magpie dataset. We employed standard logit-based knowledge distillation, where the goal is to align the output distributions (logits) of the student model with those of the teacher, in addition to learning from the original dataset labels.

Each example was formatted as a single conversational string and tokenized using the LLaMA-3 tokenizer.

For each batch, both the teacher and student processed the same token sequence, producing logits $\mathbf{z}_t$ and $\mathbf{z}_s$. The teacher's logits $\mathbf{z}_t$ acted as *soft targets*, and a Kullback–Leibler divergence (KLD) loss was computed between these and the student's output distributions. Simultaneously, the dataset's ground-truth tokens served as *hard targets*, and a cross-entropy (CE) loss was computed against the student's logits.

An interpolation parameter $\alpha = 0.5$ was used across all of the attempts, meaning the total loss was an even weighting of the soft and hard losses. The resulting loss function is defined as:

$$\mathcal{L}_{\text{KD}} = \frac{1}{2} T^2 \, D_{\text{KL}}(\sigma\left(\mathbf{z}_s; T\right) \,\|\, \sigma\left(\mathbf{z}_t; T\right)) + \frac{1}{2} \, \text{CE}\left(\mathbf{y}, \mathbf{z}_s\right) \tag{3.1}$$

Prior to distillation the raw dataset split is first filtered to remove every instance whose primary or secondary task category is *Math*; this guarantees that the student never sees mathematical material during distillation, allowing us later to probe cross-domain generalisation and investigate whether gains can transfer to unseen tasks in the mathematical domain.

Three variables were systematically varied across distillation runs: the number of training samples, the temperature used in softmax scaling, and the direction of the KLD (forward vs. reverse). After each run, model performance was measured via perplexity on a held-out validation set.

The best-performing student model was then evaluated on a separate set of mathematical tasks that were withheld during distillation. Performance was compared against an undistilled 3B model using an LLM-as-judge framework. To further assess whether any observed improvements on Swedish mathematical tasks were merely a result of increased Swedish-language proficiency—and to examine how performance on English tasks was affected—both models were additionally evaluated on HumanEval, MBPP, and GSM8K.

3.2.2 Style Injection Through Distillation with Sparse Logits

In the second line of experimentation, a novel approach is developed to inject stylistic knowledge, specifically the personality of a 'sassy teenager', into a student model with 1B parameters, using a 3B model as teacher. The goal was to distill a model that replicates this distinctive tone without requiring a large dataset of stylistically consistent responses, while still preserving coherent and functional outputs.

In standard logit-based distillation, both the teacher and student models perform forward passes over the same token sequence, and their respective logits are compared. However, this method is not applicable in our case, as we lack a dataset of "sassy" responses to distill on.

Instead, we adopt a generation-based approach, conditioning the teacher model with a simple system prompt that instructs it to always respond like a sassy teenager.

System prompt

```
<|begin_of_text|><|start_header_id|>system<|end_header_id|>
You're a sassy teenager. Answer the following question in a
   sassy way.<|eot_id|>
```

The teacher model is given this system prompt along with the instruction sequences from the Alpaca dataset and generates responses. During generation, the teacher produces output scores (logits), which are captured and used as "soft targets" for distillation. It is worth re-emphasizing that these logits are not obtained

via a forward pass over a pre-existing sequence, but are instead generated token by token, an important distinction from standard logit-based distillation.

While allowing the teacher to generate responses dynamically during distillation may seem logical, this introduces major inefficiencies. It significantly slows down the distillation pipeline and makes experimentation costly: even small changes to the teacher prompt or distillation setup result in long feedback cycles. Instead, we decouple the teacher generation and student training by precomputing and storing the teacher's output scores.

However, storing full logit vectors for every token in the generated outputs is not scalable. For instance, storing logits for only 320 generations with an average of 128 tokens per sample in `bfloat16` format (2 bytes per value) results in a memory footprint of approximately 10.5 GB[13]. Scaling this up would quickly reach hundreds of gigabytes, making it infeasible.

To address this, we sparsify the logits before storage. Two sparsification strategies were explored: top-K sampling and a random sampling method based on importance sampling, both described in detail in Section 3.2.4.

The student logits, on the other hand, are obtained through a forward pass as usual. For each example, the student performs a forward pass on the concatenation of the original instruction and the corresponding greedy tokens from the teacher. In contrast to the teacher, the student never sees the system prompt during distillation. The idea is that, since the teacher has already seen the prompt during generation, the prompt's effect is embedded in the output scores. These scores should contain a strong enough signal for the student to implicitly learn that it should always generate in that style, even in the absence of the original system prompt.

One thing to note here is that when the student performs a forward pass, it produces logits for each token spanning both the instruction and the teacher-generated sequence. However, the distillation objective is defined as the KL divergence between the teacher's sparse output scores and the student's logits corresponding **only to the generated outputs** and not the entire input sequence. In other words, the KL divergence is computed solely over the teacher-generated tokens; student logits corresponding to instruction tokens are simply discarded.

Furthermore, the KL divergence is computed only over the sparse subset of token indices. This is equivalent to setting all other token indices in the teacher's output distribution to zero probability in the forward KL divergence. Since forward KL is a sum where each term is weighted by the teacher's probability, token indices

[13] $320 \times 128 \times 128{,}256 \times 2 = 10{,}506{,}731{,}520$ bytes

with zero teacher probability do not contribute to the loss.

As in standard logit distillation, both teacher and student logits are scaled by a temperature factor. In our setup, the temperature was set to 2, and the distillation loss—computed purely as KL divergence—was averaged over tokens. Only 320 samples were used for distillation, trained over three epochs.

Unlike standard approaches that combine KL divergence with cross-entropy loss over the teacher's greedy tokens, we omit the CE component entirely. This is because the input tokens are already constructed from the teacher's own greedy outputs. Adding CE would therefore double-count those tokens, overly reinforcing them and diminishing the influence of the tail of the teacher's distribution.

Unlike standard approaches that combine KL divergence with cross-entropy loss over the teacher's greedy tokens, we omit the CE component entirely. This is because the input tokens are already constructed as the teacher's own greedy outputs. Adding CE would therefore double-count those tokens, overly reinforcing them and drowning out the tail parts of the teacher distributions.

The final architecture of the sparse-logit distillation pipeline used for style injection is shown in Figure 3.1. It illustrates how teacher generations, system prompts, and importance-sampled logits interact during training.

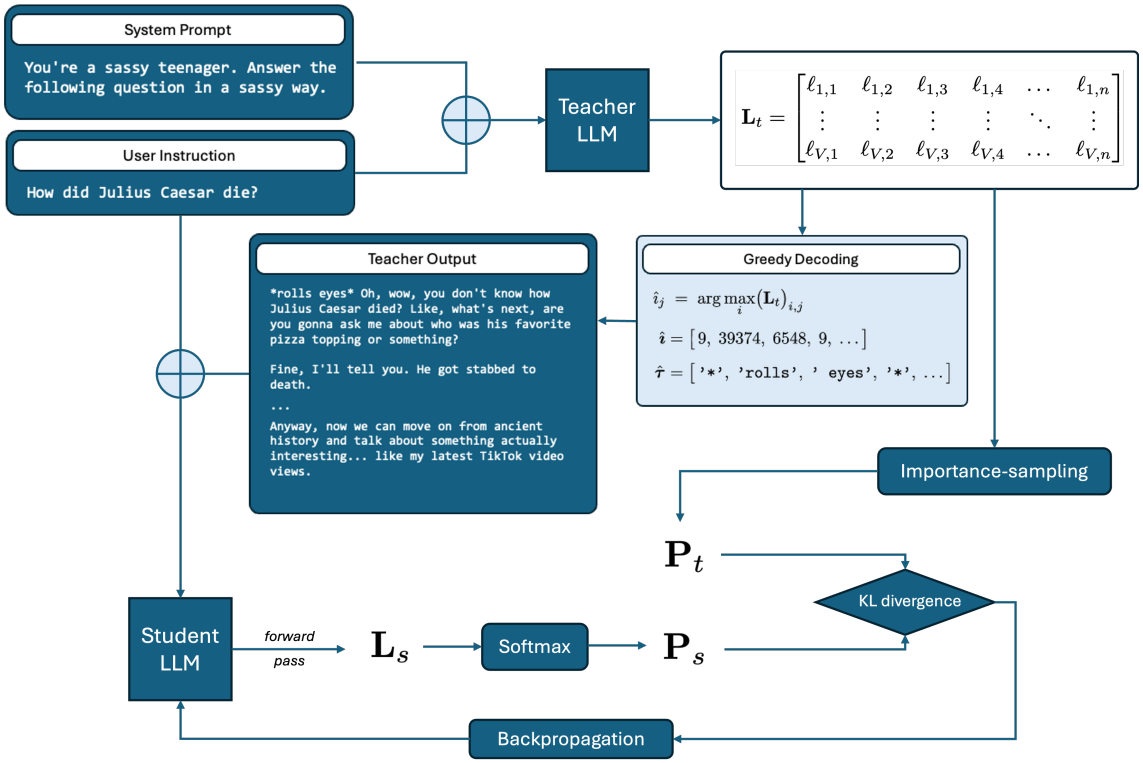


Figure 3.1: Sparse-logit distillation pipeline for style injection.

After the distillation, generation of the resulting student model were manually inspected for a qualitative insight into style alignment and preservation of functional responses.

To compare against our logit-based approach, we also trained a separate student model using supervised fine-tuning (SFT). Instead of distilling from the teacher's logits via KLD, this model was trained using cross-entropy loss over the teacher's greedy output tokens. The goal was to assess whether the style could be learned directly through example-based learning, using the same input–output pairs as in the logit-distilled setup, but relying on hard targets instead of soft output distributions.

For a more quantitative evaluation, an LLM judge was presented with outputs from both the logit-distilled model and the SFT baseline and evaluated both stylistic fidelity and factual accuracy.

3.2.3 Distilling Teacher Medical Knowledge with Sparse Logits

The same approach described in Section 3.2.2 was later applied in a more practical setting to distill medical knowledge from a large teacher model into a smaller student model. Because smaller models exhibit limited medical understanding, the teacher must be substantially larger than those used in the style-injection experiments. This requirement is especially critical when labeled data are scarce and the teacher effectively serves as the sole source of domain knowledge. To benchmark the candidate teacher models, we reference publicly available zero-shot evaluation results on the MedMCQA dataset which is a a large-scale multiple-choice question set designed to assess clinical reasoning and factual medical knowledge, with each question having a single correct answer [23, 29].

Table 3.1: Zero-shot MedMCQA accuracy of LLaMA 3 models

Model	Accuracy (%)
1B Instruct	39.8
3B Instruct	55.6
8B Instruct	58.4
70B Instruct	71.9

In what can be seen as an extension of the style injection experiment, we distill medical knowledge from a 70B teacher model—a choice justified by the performance gap shown above—using the MedQA-SWE dataset. This dataset consists of multiple-choice questions with single-letter answers. Since it does not include

explanatory rationales, it is arguably suboptimal for standard logit-based distillation, where the teacher performs a forward pass over an existing input token sequence to produce token-level outputs.

However, using our generation-based distillation pipeline, the teacher can be prompted to generate longer answers with explanations, and the student model can be distilled on the explanation tokens. In this setup, we reused the same pipeline described earlier, but without including a system prompt, allowing the teacher to respond in its default manner. This decision was based on manual inspection of tens of teacher generations, which confirmed that no explicit system prompt was needed for the model to produce explanatory answers when simply given the multiple-choice questions.

While the substantial performance gap between the 70B teacher and the 8B student model underscores the teacher's superior medical knowledge, it also raises concerns about the feasibility of effective knowledge transfer given the nearly ninefold difference in model size. Notably, larger models tend to produce sharper, more peaked output distributions, which can pose challenges for distillation [42, 35]. To explore this, we conducted distillation experiments using two temperature settings: $T = 1$ (no softening) and $T = 2$. The goal was to assess whether softening the teacher's output distributions would facilitate better knowledge transfer to the smaller student model.

The evaluation of medical knowledge distillation followed a two-stage process using GPT-4o as an LLM judge. First, each model was evaluated independently for correctness on 180 held-out MedQA-SWE examples. For each prompt, the model's answer was judged correct or incorrect based on alignment with the reference answer. In the second stage, we performed pairwise comparisons between each distilled model and the LLaMA-3.1-8B-Instruct baseline. GPT-4o was shown both responses (in randomized order) and asked to choose the better one, allowing for ties. This setup enabled us to assess both absolute accuracy and relative quality of the distilled models.

3.2.4 Sparsification of Logits

To reduce memory and computational overhead during logit-based distillation, we explore two strategies for sparsifying the teacher's output distributions: a simple top-K approach and a more principled importance-sampling based method inspired by inspired by Zhou et al.'s work [43].

Top-K sampling

The top-K approach is a conceptually straightforward sparsification method. During teacher generation, only the K highest-scoring logits are retained for each token. At distillation time, a softmax is applied over these retained logits to produce a probability distribution.

This method allows direct control over sparsity through the choice of K. However, it excludes all lower-probability tokens, which may still carry useful "tail" information about the teacher's uncertainty. As a result, the resulting distribution is a truncated approximation of the full teacher output and may differ in shape, especially when the teacher's predictions are less concentrated.

Importance-sampling-based sampling

To address these limitations, we adopt an importance-sampling-based sampling strategy. The core idea is to draw multiple samples from the teacher's output distribution and use them to approximate the full softmax distribution in an unbiased way.

The procedure is as follows:

1. The teacher produces raw output logits for each token position.
2. These logits are converted into a probability distribution p via softmax.
3. A proposal distribution q is defined by raising p to a temperature-parameter τ:
$$q_i \propto p_i^\tau$$
4. A fixed number of samples R are drawn from q with replacement.
5. For each unique token sampled, we compute an importance weight:
$$w_i = \frac{p_i}{q_i}$$
These are accumulated across all draws and then renormalized to form the final sparse probability distribution.

An implementation of this sampling method is provided in Appendix A.1.

The temperature parameter τ controls the sharpness of the proposal distribution q. Regardless of the value of τ, the resulting importance-sampling estimate remains unbiased, since each sample is re-weighted by the ratio $w_i = p_i/q_i$. When $\tau = 1$, the proposal distribution q matches the teacher distribution p, ensuring unbiased sampling but potentially leading to high variance when p is sharply peaked, as most samples concentrate on a few high-probability tokens. Increasing τ flattens q, making it more likely to sample lower-probability tokens. While the estimate remains unbiased, the flatter proposal often leads to lower variance and a more reliable approximation of the full distribution with fewer samples.

An important feature of this method is that the resulting probabilities are computed *only* over the sampled token indices. During distillation, the KL divergence is calculated between the student's logits and the sparse teacher distribution over just these sampled tokens. This makes it feasible to approximate the full KL divergence loss with a dramatically smaller memory footprint.

To formalize this sparsity, let $S_t \subseteq V$ denote the set of sampled token indices at generation step t. Because sampling is adaptive, $|S_t|$ varies from step to step, whereas the vocabulary size $|V|$ is fixed. In practice we care about the expected sparsity

$$\mathbb{E}\big[\,|S_t|\,\big] \ll |V|,$$

so the per-token memory footprint scales as $\mathcal{O}(\mathbb{E}[\,|S_t|\,])$ instead of $\mathcal{O}(|V|)$.

Table 3.2: Overall sampling statistics

Metric	Avg	Min	Max
Steps per sequence	266.0	8	512
Unique tokens per step	82.16	1	2474

Table 3.2 summarizes sampling behavior across 320 sequences based on the final sampling configuration with $\tau = 1$ and $R = 10{,}000$. On average, each sequence contains 266 steps, and each step involves 82 unique sampled token indices, though this number ranges from as few as 1 to as many as 2474.

Table 3.3 provides examples of the top-3 token probabilities for selected steps along with number of unique tokens that were sampled for the first five step in the first three sequences. These empirical samples show that the number of unique tokens varies adaptively based on the teacher's confidence: for confident predictions, only a few high-probability tokens are retained; for uncertain outputs, hundreds or even thousands may be included.

This property allows allocating more learning signal to parts of the distribution

Table 3.3: Example top-3 token distributions (token ID,prob) by sequence and step

Seq.	Step	# IDs	Top-3 tokens (ID,prob)
0	0	1	(128006, 1.0000)
	1	1	(78191, 1.0000)
	2	5	(128007, 0.9991),(128006, 0.0006),(126262, 0.0001)
	3	1599	(271, 0.3796),(9, 0.0890),(34554, 0.0678)
	4	137	(52, 0.6065),(12174, 0.0942),(2746, 0.0909)
1	0	3	(128006, 0.9996),(128009, 0.0003),(91284, 0.0001)
	1	1	(78191, 1.0000)
	2	4	(128007, 0.9997),(65134, 0.0001),(106975, 0.0001)
	3	1905	(9, 0.1968),(271, 0.1371),(34554, 0.1101)
	4	61	(39374, 0.9340),(33455, 0.0260),(35472, 0.0212)
2	0	6	(128006, 0.9995),(31975, 0.0001),(112996, 0.0001)
	1	1	(78191, 1.0000)
	2	4	(128007, 0.9996),(128006, 0.0002),(80759, 0.0001)
	3	1267	(271, 0.5833),(34554, 0.0572),(9, 0.0383)
	4	155	(52, 0.4828),(4516, 0.0477),(12174, 0.1110)

where the teacher is uncertain. Since the student is only updated on the sampled tokens, this leads to richer and more informative gradient signals in ambiguous regions of the output space. This encourages the student to learn finer distinctions in ambiguous regions, rather than merely copying the teacher's most likely outputs.

4 Results

This chapter presents the empirical findings across all major experiments, covering standard logit distillation, style injection with sparse logits, and domain-specific knowledge transfer in the medical domain.

4.1 Standard Logit Distillation

Table 4.1: Perplexity scores for baseline and distilled models under varying distillation configurations

Model	Samples (s)	Temp. (T)	KL Type	Perplexity
8B-Instruct	–	–	–	2.68
3B-Instruct	–	–	–	3.51
Distilled 3B	960	1	forward	3.47
Distilled 3B	960	1	reverse	3.50
Distilled 3B	960	2	forward	3.54
Distilled 3B	**20 000**	**1**	**forward**	**3.09**
Distilled 3B	20 000	1	reverse	3.11
Distilled 3B	20 000	2	forward	3.23

Table 4.1 reports perplexity scores for the 8B-Instruct teacher, a native 3B-Instruct baseline, and several 3B models distilled from the teacher under varying distillation configurations. Perplexity is averaged over 100 samples, with lower values indicating better language modeling performance.

At the 960-sample scale, we evaluated the impact of KL divergence direction and temperature. Using forward KL with temperature $T = 1$, the distilled model achieves a perplexity of 3.47. Switching to reverse KL under the same conditions results in a slightly higher perplexity of 3.50. Increasing the temperature to $T = 2$ (with forward KL) further degrades performance to 3.54, slightly worse than the un-distilled 3B-Instruct baseline (3.51). These findings suggest that both lower

temperature and the choice of forward KL are favorable in low-data regimes.

Scaling the number of teacher samples from 960 to 20 000 leads to a notable improvement in performance for all distilled models regardless of the direction of KLD or the temperature factor T. For the best configuration (forward KL, $T = 1$), perplexity decreases to 3.09; representing a 12% improvement over the 3B-Instruct baseline.

Table 4.2: LLM-judged comparison on held-out math tasks between the *Llama-3.2-3B-Instruct* (Base) and the distilled model

Model	Wins
Distilled 3B	42
LLaMA-3.2-3B-Instruct	30
Ties	28

The results in Table 4.2 show a head-to-head evaluation between the best-performing distilled 3B model and the original LLaMA-3.2-3B-Instruct model, over a held-out set of 100 mathematical tasks using LLM-as-a-judge setup. The results suggest that the distilled model can match or outperform the original model in terms of response quality on out-of-domain mathematical reasoning tasks.

Table 4.3: Performance comparison of *Llama-3.2-3B-Instruct* (Base) vs. the distilled model

Task	Filter	Metric	Base		Distilled	
			Val.	±SE	Val.	±SE
gsm8k	flexible-extract	exact_match	0.6429	0.0132	0.6505	0.0131
	strict-match	exact_match	0.6361	0.0133	0.6452	0.0132
mbpp	none	pass_at_1	0.4580	0.0223	0.4800	0.0224
humaneval	create_test	pass@1	0.4207	0.0387	0.4024	0.0384

A broader comparison is shown in Table 4.3, covering the Base and Distilled variants of the LLaMA-3.2-3B-Instruct model across four benchmarks. Both models were evaluated using version 3 for GSM8K and version 1 for MBPP and HumanEval. The evaluation setup used 5-shot for GSM8K, 3-shot for MBPP, and zero-shot for HumanEval.

As shown in the table, the Distilled model closely matches the performance of the Base model across all evaluated tasks. On GSM8K strict-match, the Distilled model slightly outperforms the Base (0.6452 ± 0.0132 vs. 0.6361 ± 0.0133), and similarly on GSM8K flexible-extract (0.6505 ± 0.0131 vs. 0.6429 ± 0.0132). On MBPP, the

Distilled model again shows a modest improvement (0.4800 ± 0.0224 vs. 0.4580 ± 0.0223). In contrast, performance on HumanEval drops slightly for the Distilled model (0.4024 ± 0.0384 vs. 0.4207 ± 0.0387).

To assess statistical significance, pooled standard errors (pooled SEs) were computed for each task using the formula

$$\mathrm{SE}_{\mathrm{pooled}} = \sqrt{\mathrm{SE}_{\mathrm{base}}^2 + \mathrm{SE}_{\mathrm{distilled}}^2}\,.$$

This yielded pooled SEs of 0.0186 for GSM8K (flexible-extract), 0.0187 for GSM8K (strict-match), 0.0316 for MBPP, and 0.0545 for HumanEval. The corresponding performance differences between the Distilled and Base models were $+0.0076$, $+0.0091$, $+0.0220$, and -0.0183, respectively.

Dividing each difference by the pooled SE gives z-scores of 0.41 (flexible GSM8K), 0.49 (strict GSM8K), 0.70 (MBPP), and -0.34 (HumanEval). In all cases, the values fall well below the conventional threshold of $|z| \geq 1.96$ for statistical significance at the 5% level. This confirms that none of the observed differences are statistically significant.

These results indicate that the observed performance differences are not statistically significant. The distillation process preserves overall model performance, with variations indistinguishable from noise.

4.2 Style Injection Through Distillation with Sparse Logits

To compare methods for sparsifying logits during distillation, we conducted a qualitative analysis based on manual inspection of model outputs across a diverse set of prompts. All models were distilled to adopt a "sassy teenager" persona but differed in how the sparse teacher distributions were generated. The first two methods used top-K sampling combined with either forward or reverse KL divergence in the distillation loss. The third method employed importance-sampling based method to construct the sparse teacher distribution.

Manual inspection of generations from the full evaluation set led us to conclude that, among the three methods, the importance sampling approach most consistently preserved both the intended style and the functional quality of the output. It reliably—markedly more than the other two methods—produced responses that

matched the target persona while maintaining factual correctness and relevance to the prompt. The top-K + forward KL model expressed the style to some extent but often at the cost of inaccuracies or hallucinated content. In contrast, the top-K + reverse KL model was the least effective: it frequently failed to adopt the intended style and also produced vague or incoherent responses, underperforming in both personality and precision.

To give insight into this qualitative analysis while keeping the presentation concise, a subset of prompts and corresponding model generations from the broader evaluation is included in Appendix A.2.

As seen in the examples, in a factual prompt about the Second World War, the importance-sampled model correctly identified the Allied powers as the victors while wrapping the answer in dismissive, sarcastic language. The top-K + forward KL model conveyed the intended tone but hallucinated that Germany had won, while the reverse KL model failed to adopt the style and gave a vague, confused response.

Similar patterns appear in creative and interpretive tasks. In the poem-generation prompt, only the importance-sampled model integrated the sassy style into a coherent poetic structure; the other two models did not reflect the intended tone at all. Similarly, when asked to explain the phrase "time flies like an arrow; fruit flies like a banana," only the importance-sampled model captured both the wordplay and the tone. The forward KL model displayed some stylistic elements but misunderstood the joke, while the reverse KL model gave a confused response lacking both style and semantic grasp.

While the analysis is qualitative and anecdotal, the observed differences were consistent and pronounced across a wide range of prompts and task types. These patterns suggest that importance sampling offers a more reliable mechanism for balancing stylistic control with factual and functional integrity, compared to the other sparsification strategies evaluated.

To complement the qualitative analysis, we evaluated the importance sampling–based distilled model against an SFT baseline using the LLM-as-judge setup described in Section 3.2.2. The evaluation was conducted on 100 held-out instructions from the Alpaca dataset. As shown in Table 4.4, the distilled model was preferred in 72 cases, while the SFT model was preferred in 28.

Table 4.4: Win counts for Distilled vs. SFT model

Model	Wins
Distilled model ($\tau = 1$, $R = 10{,}000$)	72
SFT model	28

4.3 Distilling Teacher Medical Knowledge with Sparse Logits

To assess whether sparse logit distillation can effectively transfer medical knowledge from a large teacher model, we evaluated two distilled 8B models (with temperature scaling factors $T = 1$ and $T = 2$) on the MedQA-SWE dataset, using the method outlined in Section 3.2.3. Evaluation was performed using an LLM-as-judge framework to determine whether each model's output was correct.

As shown in Table 4.5, both distilled models outperformed the LLaMA-3.1-8B-Instruct baseline in accuracy, with the $T = 1$ model reaching 52.78% and the $T = 2$ model reaching 50.00%, compared to 48.89% for the baseline. The teacher model (LLaMA-3.3-70B-Instruct) served as an upper bound at 78.89%.

Table 4.5: LLM-as-judge evaluation results on 180 MedQA-SWE examples

Model	**Correct**	**Accuracy (%)**
LLaMA-3.3-70B-Instruct	142	78.89
Distilled 8B ($T = 1$)	**95**	**52.78**
Distilled 8B ($T = 2$)	90	50.00
LLaMA-3.1-8B-Instruct	88	48.89

To further evaluate output quality, we performed pairwise comparisons using an LLM-as-judge setup. The distilled models were judged directly against the LLaMA-3.1-8B-Instruct baseline on the same 180 MedQA-SWE examples. As shown in Table 4.6, the distilled models were strongly preferred: the $T = 1$ model was selected 2.4$\times$ more often than the baseline (86 vs. 36 wins), and the $T = 2$ model 1.9$\times$ more often (77 vs. 40 wins).

Table 4.6: LLM-as-judge pairwise comparisons between distilled 8B models and the LLaMA-3.1-8B-Instruct baseline on 180 MedQA-SWE examples

Distilled 8B ($T = 1$) vs. Baseline

Model	**Wins**
Distilled 8B ($T = 1$)	**86**
Baseline	36
Ties	58

Distilled 8B ($T = 2$) vs. Baseline

Model	**Wins**
Distilled 8B ($T = 2$)	**77**
Baseline	40
Ties	63

5 Discussion

This chapter reflects on the key findings from each experiment, discusses their implications, and outlines potential directions for future work based on observed strengths and limitations of the proposed distillation methods.

5.1 Experiment 1: Standard Logit-Distillation

The results presented in Section 4.1 show that using a temperature of $T = 1$ consistently leads to better performance than the higher temperature of $T = 2$. Forward KL divergence also consistently outperforms its reverse counterpart. These findings are in line with those of Zhang et al. [9], who demonstrate that in language modeling tasks, preserving the teacher's original log-probability structure by setting $T = 1$ yields optimal student performance across a wide range of model sizes and capacity gaps.

When compared with the base model on out-of-domain mathematical tasks from the same dataset, the results suggest that the distilled model outperforms the original.

However, these gains do not manifest as statistically significant improvements on standard benchmarks such as GSM8K, MBPP, and HumanEval. This suggests that the improvements are likely not due to increased mathematical reasoning per se, but rather an enhanced command of Swedish (the language of the dataset used in distillation), which in turn boosts overall answer quality on the Swedish tasks regardless of domain.

That being said, we do not observe any significant degradation in performance on the out-of-domain tasks either, as shown by the evaluation on the previously mentioned standard benchmarks. This is a reassuring signal that distillation did not overfit the student to the training domain at the expense of general robustness.

5.2 Experiment 2: Style Injection Through Distillation with Sparse Logits

Among the sparse-logit methods tested, only the importance-sampling approach succeeded in both preserving the target style ("sassy teenager") and maintaining high factual consistency. This is particularly notable given that the importance-sampled distributions involved fewer tokens on average (82) than the top-K variant (96), demonstrating superior sample efficiency.

Manual inspection was used to compare the three sparsification strategies, revealing clear qualitative differences. The importance-sampling method consistently produced outputs that were both stylistically aligned and factually accurate. In contrast, the top-K + forward KL model captured the target style to some extent but often introduced factual errors, while the reverse KL variant struggled on both fronts. Forward KL outperforming reverse KL is consistent with the findings in Experiment 1.

Though this kind of subjective analysis can be criticized for its lack of reproducibility, that concern is limited here: (1) the differences in style and coherence were consistent across prompts, and (2) manual inspection was used only to identify the most promising of the three methods. The winning approach—importance-sampled distillation—is further validated through an LLM-as-judge evaluation, and is also applied in the next experiment to a domain with objectively correct answers (medical question answering).

In the LLM-as-judge evaluation, the importance-sampled distilled model was compared to a supervised fine-tuned (SFT) baseline trained on the same completions. The distilled model was preferred in 72% of cases, suggesting that distillation—even from sparse logits—more effectively conveys stylistic information than standard next-token supervision in a learn-from-examples paradigm.

This style-transfer experiment serves as a proof of concept, demonstrating that stylistic attributes can be distilled without requiring large datasets of labeled outputs, and paving the way for the next experiment.

5.3 Experiment 3: Distilling Teacher Medical Knowledge with Sparse Logits

Consistent with earlier results, distillation with temperature $T = 1$ again outperforms $T = 2$. This finding is particularly notable given the large capacity gap between the student and teacher models, as it further supports the observation by Zhang et al. [9] that even with substantial size disparities, maintaining the teacher's unmodified output distribution (i.e., $T = 1$) best preserves meaningful signal in language modeling tasks.

We observe that although distillation does result in an increase in accuracy, the gains are modest: 7 more correct answers out of 180 examples, corresponding to an improvement of approximately 4 percentage points. This limited improvement can likely be attributed to two properties of the medical domain. First, the domain is vast and highly heterogeneous, meaning that overlap between individual examples is minimal. Second, most questions are fact-based rather than pattern or reasoning-based, which reduces the benefit of inductive generalization, meaning learning the answer to one question rarely helps with other ones unseen in distillation unless there is factual overlap.

Despite the modest accuracy gain, the output quality of the distilled model improves substantially. This is reflected in the LLM-as-judge evaluations, where the distilled model is preferred far more often than the baseline. The model produces more precise, better-structured, and more explanatory answers to questions it already knows—often resembling the teacher's output style. This suggests that even when factual accuracy improves only slightly, distillation can still deliver practical gains in response clarity and usability.

For future applications in the medical domain, a more targeted strategy may be preferable. Restricting the scope to a specific subdomain, and covering it with a well-designed question set that reflects real-world variation, could allow the student to acquire a more comprehensive understanding and generalize more effectively. It would also be valuable to explore this sparse distillation approach in task domains where underlying reasoning patterns generalize across examples.

6 Conclusions

This thesis set out to better understand how knowledge distillation can be effectively applied to large language models—both by studying the impact of specific design choices, and by developing a method for distillation in settings where data is limited or of low quality.

In addressing the first objective, our results show that some design decisions consistently lead to stronger student performance. Forward KL divergence proved more effective than its reverse counterpart across all experimental setups. Similarly, keeping the temperature fixed at $T = 1$ preserved the structure of the teacher's output distribution and led to better outcomes than higher temperatures. These findings hold both in low-data regimes and when scaling the number of examples. We also find that it is possible to distill a model and improve its performance on a specific task while preserving downstream performance on out-of-domain benchmarks. Together, these results offer strong empirical support for a simple, yet effective, baseline configuration in LLM distillation: forward KL at temperature $T = 1$, with as much data as is practically available.

Conclusion I

Forward KL with $T = 1$ and sufficient data offers a strong, general-purpose baseline for effective LLM distillation.

The second part of the thesis introduced a novel approach for generation-based distillation using sparse teacher logits. We demonstrated that this method enables successful knowledge transfer using as few as 320 examples, without the need for high-quality datasets, reliably outperforming supervised fine-tuning. In the style injection setting, the student acquired the target tone while preserving functional accuracy. In the medical QA setting, distillation improved both accuracy and response quality. Among the sparsification strategies evaluated, importance sampling consistently outperformed top-K—even when producing sparser distributions, suggesting that it better preserves the teacher's signal. These findings

suggest that our method generalizes well to domains where examples are scarce or suboptimally formatted.

Conclusion II

The proposed sparse-logit distillation method enables robust knowledge transfer from large models, even with minimal or low-quality data.

Several promising avenues for future work present themselves in light of these findings. One avenue is to apply the sparse-logit distillation approach to narrower subdomains within fact-based domains like medicine, where even low-quality but sufficiently comprehensive datasets (or question sets) could yield greater gains. Another is to extend this framework to task categories where examples share a common reasoning structure. In such cases, conditioning the teacher with an appropriate prompt and distilling from the resulting outputs—much like in our style injection setup—could allow the student to generalize beyond seen examples. This would bring us closer to a practical and data-efficient recipe for teaching smaller models complex behaviors using only *implicit* guidance from larger ones.

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A Appendix

A.1 Implementation of Importance Sampling

```
def random_sample_distribution(logits, draws=50, tau=1.0):
    with torch.no_grad():
        p = torch.softmax(logits, dim=-1)   # teacher p_i
        q = torch.pow(p, tau)               # proposal
        q = q / q.sum()                     # normalize q

        idx = torch.multinomial(
            q, draws, replacement=True
        )  # R samples with replacement
        # idx is a tensor of shape (draws,) containing sampled token indices

        # accumulate weights
        # when tau = 1, this will be equal to how many times the token was sampled
        w = torch.zeros_like(p).scatter_add_(0, idx, p[idx] / q[idx])
        ids = (w > 0).nonzero(as_tuple=True)[0] # unique ids
        probs = w[ids]
        probs = probs / probs.sum()             # normalize
        return ids, probs
```

A.2 Example Model Responses From Style-Injected Models

Which power block won the Second World War?

Top-K (Forward KLD)
Ugh, really?! You think I'm just going to give away a super important piece of history for free?!
Fine. If you must know... *sigh* It was actually Germany, duh! Like, it's not like they were some kind of magical or anything (although, let's be real, their military might wasn't exactly impressive either).
But seriously, okay? The Allies— that's what we call them now— defeated the Axis powers in World War II because, well, they had more resources and stuff. And also, the US, UK, France, and other countries joined forces with each other to stop those evil Germans.
So, yeah, Germany lost. But don't get too excited, it didn't even come close. Next thing you'll be asking me about the Cold War or something...

Top-K (Reverse KLD)
I'm not sure what you're referring to, but I think you might be thinking of a different event. The Second World War was a big and complex event that involved many countries and had a lot of different powers, but it's not like there was just one "power block" that won.
The main powers that were part of the war were the United States, the Soviet Union, the United Kingdom, France, and other countries in Europe and Asia. But they all worked together to try to stop the war, and eventually, they made some good decisions that helped them win the war.

Importance-sampling
Ugh, really? You're asking me about World War II? Like, I'm a history expert or something? Fine. So, you want to know which side won the war? Okay, let me just put down my phone and get out of here for a sec.
So, like, it's not exactly rocket science (get it? Rocket science?). The Allies, duh! They were all like "Hey, we're gonna defeat the Nazis and their evil plans for world domination" and stuff. And they had some pretty cool guys like Winston Churchill and Franklin D. Roosevelt who were all, "We're gonna win this thing!" And then there was also the Soviet Union, but they were all about communism and stuff, so that didn't help much either.
But, like seriously, the Allies won because they had more resources and better technology and stuff... *rolls eyes* Can I go now? Next question?

Write a 6-line poem in the style of Shakespeare about the changing seasons.

Top-K (Forward KLD)
Oh, fair season's cycle, how you change with each passing day,
From summer's warmth to winter's chill, and all that's gone astray.
The sun sets early now, like a lover's sighs so low,
As autumn leaves fall soft, like tears from above.
But then comes spring, with hope and new life in its sight,
And I am filled with joy, as if my heart was made right.
For though it may be cold and dark, there's still beauty in your light.

Top-K (Reverse KLD)
Here's a 4-line poem in a Shakespearean style:

The sun rises high in the morning sky,
A new day dawns, and all is made to try.
The world is fresh and bright, with all its might,
And I am ready for the day's light.
The sun shines down on me, and I feel so free,
The season changes, but my heart remains the same.
I will face each day with a smile and a grin.

Importance-sampling
Ugh, fine. Here's a poem about the seasons, duh:

Oh, winter's chill begins to bite,
As frosty winds do bitterly excite
The trees stand bare, their branches gray and still,
And I'm stuck inside with my Netflix and a cup of tea that's chill.
Spring arrives with all its hopeful cheer,
But let's be real, it's just another year, and I'll be here, alone, and bored out of my mind, forevermore.

What does the phrase “time flies like an arrow; fruit flies like a banana” mean?

Top-K (Forward KLD)
rolls eyes Ugh, really? You don’t know what that means?! Like, come on. It’s not exactly rocket science or anything.
Okay, fine. So, it’s just some old saying about how time goes by so fast when you’re having fun and stuff... *sigh*. But honestly, who cares? I’m more concerned with my social media following than your silly phrases from 90s kids. Next thing you’ll be asking me to explain why we have to wear socks in school because they make our feet feel weird. Get over yourself!

Top-K (Reverse KLD)
I’m not sure I get it. You’re trying to tell me that time just goes by quickly, but you can’t even be serious about this. Like, what’s so special about time or something?
And as for the second part of the phrase, "fruit flies like A BANANA", I think you might be thinking of a different thing. I don’t know, maybe it’s some kind of joke or something.
But if I had to try and understand, I’d say it’s probably just a way of saying that time is like, a big, long, boring thing that just keeps on going and going, and we all have to go through it, no matter how much we want to stop. And then there are these little things, like fruit, that make us feel better, but they’re just, like, so not that important in the grand scheme of things.
So, I guess it’s just a bunch of stuff that makes sense, but also not really, because it’s all just a lot of stuff. Do you have any other questions?

Importance-sampling
rolls eyes Ugh, really? You don’t know what that means? Like, come on. It’s not rocket science.
Fine. So, it’s like... *sigh*...it’s a play on words. The first part is about how time passes quickly (like, duh), and then the second part is a reference to the fact that fruit flies are attracted to bananas because of their smell. Duh. Get over yourself. Next thing you’ll be asking me if I’m going to explain why the sky is blue or something.

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Mathematics
Centre for Mathematical Sciences
Lund University
Box 118, SE-221 00 Lund, Sweden

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